

CE807 · Text Analytics

School of Computer Science and Electronic Engineering - University of Essex

Electronic Submission and check submission date and time on FASER (<https://faser.essex.ac.uk/>)

Please also see your student handbook for rules regarding the late submission of assignments

On Plagiarism

The work you submit must be your own. Any material you use, whether it is from textbooks, classmates, the web, or any other source, must be acknowledged in your work.

All submissions are fairly and transparently checked for plagiarism. Please make sure that you provide frequent citations. But also make sure that each sentence written is originally yours, i.e., the material is read, understood, and the report is written using your own words and your own language only. Do not copy and paste and rephrase the copied text.

There are many different forms of what is considered plagiarism. For example, based on the feedback from the SAO officer, many students were unaware that, e.g., copying entire paragraphs without clearly identifying them as quotes, etc. is a form of plagiarism, etc. Thus, please check back with your scientific writing module before you submit it! Please see the university's academic integrity [policy](#).

Further note that also plainly reusing software code or merely slightly adapting existing software code and submitting it as one's own fulfils the matter of plagiarism. Cite any code that you reuse, too.

In previous years, ~20% of the submitted reports were plagiarised. There were also multiple cases of software code plagiarism. This number is too high and shall be 0% in this year!!!

The assessment has two parts

- **Moodle Quiz (30%):** <https://moodle.essex.ac.uk/mod/quiz/view.php?id=1453054>
- **Assignment (70%):** Assignment briefing below (**Submission via Faser**)

Motivation: Text classification is a crucial task in **Text Analytics** that underpins various applications such as spam filtering, sentiment analysis, document categorization, and automated content moderation. With the increasing volume of digital text data, the ability to classify and organize information efficiently has become essential across industries, including finance, healthcare, education, and media.

This assignment aims to develop your understanding of applying traditional machine learning algorithms, specifically **MLP** and **Zero-shot LLM Based Prompting using open source LLM** for **Text Classification** tasks. This assignment is designed to provide practical experience in implementing text classification models using two models one traditional and another LLM based. By engaging with real-world datasets, you will develop expertise in data pre-processing, feature engineering, model training, and performance evaluation. Beyond technical proficiency, this assignment encourages critical thinking in evaluating model performance, addressing challenges like class imbalance, interpretability, and domain-specific nuances. For logistics regression, you will use **different pre-processing techniques** and finalize the performing techniques. For LLM Based prompting, you will be at least **5 different prompts** to find out which prompt works best and why.

Code: You will be provided with a sample code and its structure (https://drive.google.com/file/d/1YOuq_21wE3GUZHVFcyDMQdGUiRDp1u/view?usp=sharing) . You must follow the same structure for coding purposes. Note the code provided in a sample code, you need to modify it appropriately including names. Your code will be automatically executed, so please follow the instructions to receive the mark. You are allowed to use codes from the lab or web, provided you include proper citations and demonstrate an understanding of the code. You need to modify the code to fit the structure provided. You must ensure that your code is runnable with one click; any special package requirements need to be automatically installed.

Dataset: https://drive.google.com/file/d/1kGCZkCXu95MYMimnj_XIQJyb9vDAuXrh/view?usp=sharing

Output: You must save the model's output in the "test.csv" by adding/modifying "out_label_MLP", "out_label_Prompt_1", "out_label_Prompt_2", "out_label_Prompt_3", "out_label_Prompt_4" and "out_label_Prompt_5" columns in the existing columns from test.csv. Note that you shouldn't change any other column name in the "test.csv" file. You must use the same column names; otherwise, the automatic code will not work and will not be marked. You will also save models over the web (such as Google Drive) so that they can be automatically accessed.

Report & Presentation: You must submit the report in the ACL format [Template](#) with the same structure as provided. The link to the report format will be provided on the Moodle page. **You must walk through your code in a 15-minute video presentation.** Your face, and audio must be present in the video presentation. You must use Zoom to record the presentation.

Submission: The assignment must be submitted in the following files having the same names and no other files

- **Assignment: Report & Presentation**

- **report.pdf:** Your report file with video presentation link
- **Assignment: Coding and Output**
 - **code.ipynb:** Your code as a single file
 - **test.csv:** Your output for both models
- **Note all 3 files need to be submitted via Faser link provided.**

TASKS

This assignment counts towards 70% of the overall mark for CE807 and is divided into two parts having equal weightage:

- **Assignment: Report & Presentation (50%)**
 - Task 1: Model Discussion (25%)
 - Task 4: Discussion & Summary (25%)
 - **Submission**
 - **report.pdf:** Your report file with video presentation links
- **Assignment: Coding and Output (50%)**
 - Task 2: Design, implementation of Model (25%)
 - Task 3: Output Comparison and Analysis (25%)
 - **Submission**
 - **code.ipynb:** Your code as a single file
 - **test.csv:** Your output for both models

Task 1: Model Discussion (25%)

Whenever you develop a new classifier or other text analytics software, you must select the best techniques. In this assignment, you must select two different Models: **MLP and Zero-shot LLM Based Prompting (atleast 5 different prompts on open source LLM)**. You need to briefly summarize both models and it's intermediate steps like pre-processing, prompts design, validation set use etc.

A 25% weightage is given to testing different open source LLMs and finding best one to use with explanation.

Task 2: Design, implementation of classifiers (25%)

For this task, you will develop classifier models based in discussed Task 1. Doing this will involve one or more of the following steps:

- Identifying the approaches and/or features you want to extract.
- Apply different Pre-processing techniques to the text based on the dataset and model requirements.
- Developing code to extract text representation
- Train selected model classifiers.
- Experimentally justify different pre-processing and text representations.
- Use validation set and/or parameter search to find the optimal model setting.

- Evaluate the performance of the classifier using a scientifically sound methodology.
- Select Open source LLM from Hugging face.
- Prompting different LLMs and selecting best one
- Post-processing LLM outputs

A 25% weightage is assigned for systematically using models on various parameters to select the best models using a validation set, as well as for producing properly documented codes.

Task 3: Output Comparison and Analysis (25%)

The goal of the task is to explain and analyze the model's performance. You need to justify and compare the performance of both models. Justification must contain both theoretical and practical explanations. You will also select **at least 5 diverse and interesting examples**, provide their ground truth/true label, and compare their model across models. You must compare why model's output is correct or wrong. You can also examine the model's confidence and provide a comment on it.

A 25% weightage is assigned for systematically comparing the performance of both models at a granular level and connecting example outputs with theoretical or experimental explanations.

Task 4: Discussion and Summary (25%)

Finally, you will write a report documenting your actions and the findings. You should explain why you chose the algorithms and features you used and how they compare to state-of-the-art approaches. You should discuss the performance of your approach and reflect on what you have learned. Simply stating that you've learned a new model, coding, etc., is not sufficient. Your report should be **4-5 pages** in length, excluding references, formatted in the ACL style.

You will also prepare a Zoom recorded the presentation for **12-15min** (not more than 15min) with **your face visible**. You must use the university's [zoom](#), save your presentation on the university cloud, and provide a shareable link to the recorded video in the report. In the presentation, you will **walk through the code** and explain the main building blocks of the code. You will show testing code and models producing outputs during the presentation. You can preload models, but you must run the test examples while recording.

Video Presentation carries 25% weightage. Therefore, if something is missing from your video, for example, if your presentation video lacks your voice, face, or running code, you will lose 25% of the marks. It is your responsibility that your video works as required.

Extenuating circumstances or Late of Submission

For any reason, you are late in your assignment submission. Please don't contact me. There is university wise guideline for this, you should follow that. In general, you should read your student [handbook](#) for more details.

- Making an extenuating circumstance claim
 - <https://www.essex.ac.uk/student/exams-and-coursework/extenuating-circumstances>
- Late submission of coursework
 - <https://www.essex.ac.uk/student/exams-and-coursework/late-submission-of-coursework>

If you are in any queries or need some advice, please contact csee-assessment@essex.ac.uk or the school Office team for further advice.

Again, please don't contact me regarding EC.

Plagiarism Policy

In case you are reported for suspected academic offences. You will get information from the school office on how to proceed next. Suspected academic offences don't mean you will have a penalty. A team will be involved and decide the next course of action in which I am not involved. **Please don't contact me regarding suspected academic offences (SAO).** At every step, the school office will keep you informed. In the meantime, you could look at the [University's guidance and procedures about academic offences](#) and understand beforehand what will happen at the meeting. If you think you need some support or guidance before the meeting, you get in touch with [SU Advice](#), who provide free, expert, and independent advice to all students.

The wait time or the process could be stressful. Please get in touch with the [University's student wellbeing service](#), which provides confidential advice and assistance.

Re-marking Policy

It's important to note that being dissatisfied with your mark is not grounds for requesting a remark. You can find the grounds on which you can appeal a mark in section 3 of the marking policy. You can review the marking policy at <https://www.essex.ac.uk/student/exams-and-coursework/assessment-and-marking-policies>

In minor cases, some feedback might not match the exact submission. For example, in "Training of classifier using selected models," feedback says, "no use of validation set," but you have used a validation set. This makes a case for a relook. You must provide all such feedback, and only those will be relooked. However, that might have a minor or no impact on overall marks because marking is done by looking at everything together, and wrong feedback was selected. Please mail me for these cases with the email subject "**CE807-26-SP: Assignment (1234567)**", where 1234567 is your 7/8 digits student id.

MARKING BREAKDOWN (out of 100%)

Task 1: Tokenization, Representation & Discussion (25%)

- Summary of both models **(up to 10%)**
- Summary of both model's parameters **(up to 05%)**
- Critical discussion and justification of parameters **(up to 10%)**
- 25% weightage for testing different LLMs and finding best one to use.

Task 2: Design and implementation of classifiers (25%) - Previous required

- Training & Testing of both models **(up to 15%)**
- High-quality code including comments and printing required measures etc **(up to 10%)**
- A 25% weightage is assigned for systematically using models on various parameters to select the best models using a validation set, as well as for producing properly documented codes.

Task 3: Output Comparison and Analysis (25%) - Previous required

- Comparison of both model's performance **(up to 10%)**
- Providing Justification of Model performance **(up to 05%)**
- Diverse and Interesting Example Selection **(up to 10%)**
- 25% weightage is for systematically comparing both selected models' performance at a granular level and connecting example output with a theoretical or experimental explanation.

Task 4: Discussion & Summary (25%) - Previous required

- Explanation of selected examples and model output **(up to 10%)**
- Lessons Learned **(up to 05%)**
- Material submitted in appropriate format **(up to 10%)**
- 25% weightage is for video presentation.

Other Points

- **No report submission**
 - Submission without a report could get you 50% maximum marks
- **No Workable Code submission**
 - Submission without a code could get you 50% maximum marks.